

28.1 Features

- Internally resets chip if not periodically reloaded.
- Debug mode.
- Enabled by software but requires a hardware reset or a Watchdog reset/interrupt to be disabled.
- Incorrect/Incomplete feed sequence causes reset/interrupt if enabled.
- Flag to indicate Watchdog reset.
- Programmable 32-bit timer with internal pre-scaler.
- Selectable time period from $(T_{WDCLK} \times 256 \times 4)$ to $(T_{WDCLK} \times 2^{32} \times 4)$ in multiples of $T_{WDCLK} \times 4$.
- The Watchdog clock (WDCLK) source can be selected from the Internal RC oscillator (IRC), the APB peripheral clock (PCLK, see [Table 40](#)), or the RTC oscillator. This gives a wide range of potential timing choices for Watchdog operation under different power reduction conditions. For increased reliability, it also provides the ability to run the Watchdog timer from an entirely internal source that is not dependent on an external crystal and its associated components and wiring.
- The Watchdog timer can be configured to run in Deep Sleep mode when using the IRC as the clock source.

28.2 Applications

The purpose of the Watchdog is to reset the microcontroller within a reasonable amount of time if it enters an erroneous state. When enabled, the Watchdog will generate a system reset if the user program fails to "feed" (or reload) the Watchdog within a predetermined amount of time.

For interaction of the on-chip watchdog and other peripherals, especially the reset and boot-up procedures, please read [Section 3.4 "Reset" on page 20](#) of this document.

28.3 Description

The Watchdog consists of a divide by 4 fixed pre-scaler and a 32-bit counter. The clock is fed to the timer via a pre-scaler. The timer decrements when clocked. The minimum value from which the counter decrements is 0xFF. Setting a value lower than 0xFF causes 0xFF to be loaded in the counter. Hence the minimum Watchdog interval is $(T_{WDCLK} \times 256 \times 4)$ and the maximum Watchdog interval is $(T_{WDCLK} \times 2^{32} \times 4)$ in multiples of $(T_{WDCLK} \times 4)$. The Watchdog should be used in the following manner:

- Set the Watchdog timer constant reload value in WDTC register.
- Setup the Watchdog timer operating mode in WDMOD register.
- Enable the Watchdog by writing 0xAA followed by 0x55 to the WDFEED register.
- The Watchdog should be fed again before the Watchdog counter underflows to prevent reset/interrupt.

When the Watchdog is in the reset mode and the counter underflows, the CPU will be reset, loading the stack pointer and program counter from the vector table as in the case of external reset. The Watchdog time-out flag (WDTOF) can be examined to determine if the Watchdog has caused the reset condition. The WDTOF flag must be cleared by software.

The watchdog timer block uses two clocks: PCLK and WDCLK. PCLK is used for the APB accesses to the watchdog registers. The WDCLK is used for the watchdog timer counting.

There is some synchronization logic between these two clock domains. When the WDMOD and WDTC registers are updated by APB operations, the new value will take effect in 3 WDCLK cycles on the logic in the WDCLK clock domain. When the watchdog timer is counting on WDCLK, the synchronization logic will first lock the value of the counter on WDCLK and then synchronize it with the PCLK for reading as the WDTV register by the CPU.

28.4 Register description

The Watchdog contains 4 registers as shown in [Table 523](#) below.

Table 523. Watchdog register map

Name	Description	Access	Reset Value ^[1]	Address
WDMOD	Watchdog mode register. This register contains the basic mode and status of the Watchdog Timer.	R/W	0	0x4000 0000
WDTC	Watchdog timer constant register. This register determines the time-out value.	R/W	0xFF	0x4000 0004
WDFEED	Watchdog feed sequence register. Writing 0xAA followed by 0x55 to this register reloads the Watchdog timer with the value contained in WDTC.	WO	NA	0x4000 0008
WDTV	Watchdog timer value register. This register reads out the current value of the Watchdog timer.	RO	0xFF	0x4000 000C
WDCLKSEL	Watchdog clock source selection register.	R/W	0	0x4000 0010

[1] Reset Value reflects the data stored in used bits only. It does not include reserved bits content.

28.4.1 Watchdog Mode register (WDMOD - 0x4000 0000)

The WDMOD register controls the operation of the Watchdog as per the combination of WDEN and WRESET bits. Note that a watchdog feed must be performed before any changes to the WDMOD register take effect.

Table 524: Watchdog Mode register (WDMOD, address 0x4000 0000) bit description

Bit	Symbol	Description	Reset Value
0	WDEN	WDEN Watchdog enable bit (set-only). When 1, the watchdog timer is running.	0
1	WRESET	WRESET Watchdog reset enable bit (set -only). When 1, a watchdog time-out will cause a chip reset.	0
2	WDTOF	WDTOF Watchdog time-out flag. Set when the watchdog timer times out, cleared by software.	0 (Only after POR)
3	WDINT	WDINT Watchdog interrupt flag (read-only, not clearable by software).	0
31:4	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

Once the **WDEN** and/or **WRESET** bits are set they can not be cleared by software. Both flags are cleared by an external reset or a Watchdog timer underflow.

WDTOF The Watchdog time-out flag is set when the Watchdog times out. This flag is cleared by software.

WDINT The Watchdog interrupt flag is set when the Watchdog times out. This flag is cleared when any reset occurs. Once the watchdog interrupt is serviced, it can be disabled in the NVIC or the watchdog interrupt request will be generated indefinitely. the intent of the watchdog interrupt is to allow debugging watchdog activity without resetting the device when the watchdog overflows.

Watchdog reset or interrupt will occur any time the watchdog is running and has an operating clock source. Any clock source works in Sleep mode, and the IRC works in Deep Sleep mode. If a watchdog interrupt occurs in Sleep or Deep Sleep mode, it will wake up the device.

Table 525. Watchdog operating modes selection

WDEN	WRESET	Mode of Operation
0	X (0 or 1)	Debug/Operate without the Watchdog running.
1	0	Watchdog interrupt mode: debug with the Watchdog interrupt but no WRESET enabled. When this mode is selected, a watchdog counter underflow will set the WDINT flag and the Watchdog interrupt request will be generated.
1	1	Watchdog reset mode: operate with the Watchdog interrupt and WRESET enabled. When this mode is selected, a watchdog counter underflow will reset the microcontroller. Although the Watchdog interrupt is also enabled in this case (WDEN = 1) it will not be recognized since the watchdog reset will clear the WDINT flag.

28.4.2 Watchdog Timer Constant register (WDTC - 0x4000 0004)

The WDTC register determines the time-out value. Every time a feed sequence occurs the WDTC content is reloaded in to the Watchdog timer. It's a 32-bit register with 8 LSB set to 1 on reset. Writing values below 0xFF will cause 0x0000 00FF to be loaded to the WDTC. Thus the minimum time-out interval is $T_{WDCLK} \times 256 \times 4$.

Table 526: Watchdog Constant register (WDTC, address 0x4000 0004) bit description

Bit	Symbol	Description	Reset Value
31:0	Count	Watchdog time-out interval.	0x0000 00FF

28.4.3 Watchdog Feed register (WDFEED - 0x4000 0008)

Writing 0xAA followed by 0x55 to this register will reload the Watchdog timer with the WDTC value. This operation will also start the Watchdog if it is enabled via the WDMOD register. Setting the WDEN bit in the WDMOD register is not sufficient to enable the Watchdog. A valid feed sequence must be completed after setting WDEN before the Watchdog is capable of generating a reset. Until then, the Watchdog will ignore feed errors. After writing 0xAA to WDFEED, access to any Watchdog register other than writing 0x55 to WDFEED causes an immediate reset/interrupt when the Watchdog is enabled. The reset will be generated during the second PCLK following an incorrect access to a Watchdog register during a feed sequence.

Table 527: Watchdog Feed register (WDFEED, address 0x4000 0008) bit description

Bit	Symbol	Description	Reset Value
7:0	Feed	Feed value should be 0xAA followed by 0x55.	NA
31:8	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

28.4.4 Watchdog Timer Value register (WDTV - 0x4000 000C)

The WDTV register is used to read the current value of Watchdog timer.

When reading the value of the 32-bit timer, the lock and synchronization procedure takes up to 6 WDCLK cycles plus 6 PCLK cycles, so the value of WDTV is older than the actual value of the timer when it's being read by the CPU.

Table 528: Watchdog Timer Value register (WDTV, address 0x4000 000C) bit description

Bit	Symbol	Description	Reset Value
31:0	Count	Counter timer value.	0x0000 00FF

28.4.5 Watchdog Timer Clock Source Selection register (WDCLKSEL - 0x4000 0010)

This register allows selecting the clock source for the Watchdog timer. The possibilities are the Internal RC oscillator (IRC) or the APB peripheral clock (pclk). The function of bits in WDCLKSEL are shown in [Table 529](#). The clock source selection can be locked by software, so that it cannot be modified. On reset, the clock source selection bits are always unlocked.

When the IRC is chosen as the watchdog clock source, the watchdog timer can remain running in deep sleep mode, and can reset or wake up the device from that mode.

Table 529: Watchdog Timer Clock Source Selection register (WDCLKSEL, address 0x4000 0010) bit description

Bit	Symbol	Value	Description	Reset Value
1:0	WDSEL		These bits select the clock source for the Watchdog timer as described below. Warning: Improper setting of this value may result in incorrect operation of the Watchdog timer, which could adversely affect system operation. If the WDLOCK bit in this register is set, the WDSEL bit cannot be modified.	0
		0x0	IRC. Selects the Internal RC oscillator (irc_clk) as the Watchdog clock source (default).	
		0x1	PCLK. Selects the APB peripheral clock (watchdog pclk) as the Watchdog clock source.	
		0x2	RTC. Selects the RTC oscillator (rtc_clk) as the Watchdog clock source.	
		0x3	Reserved, this setting should not be used.	
30:2	-	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
31	WDLOCK		WDT locking.	0
		0	Unlocked. This bit is set to 0 on any reset. It cannot be cleared by software.	
		1	Locked. Software can set this bit to 1 at any time. Once WDLOCK is set, the bits of this register cannot be modified.	

28.5 Block diagram

The block diagram of the Watchdog is shown below in the [Figure 133](#). The synchronization logic (PCLK - WDCLK) is not shown in the block diagram.

